

Dynamic Traffic Signal Control using Auction and Negotiation Based Protocols on the Edge

Team Ohmies

Abstract

We propose a decentralized traffic control system where each intersection is an autonomous agent with capabilities for local vehicle sensing, heuristic based signal control, and negotiation with its peers. To achieve this, we integrate and extend concepts from two key research papers: the predictive communication model from Jiang et al. [1] to enable agents to forecast future traffic, and the auction based mechanism to determine phase changes inspired by Baluja et al. [2]. Our primary contribution is the design of a negotiation protocol that links these two concepts, allowing agents to cooperatively manage traffic flow by exchanging congestion reduction requests. The protocol is designed with authentication and resilience mechanisms to ensure the safe operation of the agents, and graceful degradation during network disruptions.

1 Introduction

This proposal outlines a protocol to address the requirements of the project, where the core task is to design and implement a lightweight, heuristic-based negotiation protocol that allows agents to arbitrate green signal allocation when traffic builds up. Our approach is to develop a system where agents negotiate proactively, based on predictions of future traffic conditions, as well as the current queue and wait time. This requires a combination of predictive modeling and dynamic arbitration, for which we draw upon established research from [1] and [2], as described in the following section.

2 Background and Related Work

2.1 Predictive Communication for Multi-Agent Coordination

In multi-agent traffic control, effective coordination requires communication, but naive information sharing can be detrimental. The work by Jiang et al. in [1] addresses the problem of information overload, where agents sharing large, raw observational data can destabilize and slow down the process. The authors posit that an agent does not need to know its neighbor's complete state; it only needs to know the *impact* of that neighbor's actions on itself. The most direct impact is the volume of traffic one intersection sends to another. The proposed solution, UniComm, is a lightweight communication protocol where agents use a neural network to predict the number of vehicles they will send to each neighbor. This single, predictive scalar value is the only information exchanged, making the protocol minimal, efficient, and directly relevant for coordination.

Our protocol adapts this core principle to satisfy the requirement of being a lightweight protocol. Each of our agents will use a predictive model to generate a forecast of its outgoing traffic, and send it to neighboring agents for their use.

2.2 Auction-Based Mechanisms for Traffic Control

Traditional traffic signals with fixed timers are inefficient because they are unresponsive to real-time demand. The work by Baluja et al. in [2] re-frames the problem of green time allocation as an internal auction. Each traffic movement bids for the right to use the intersection, with the bid value representing its urgency. At frequent intervals, a micro-auction is held, and the movement with the highest bid wins the green signal. Bids are calculated from real-time sensor data, creating a highly adaptive system that naturally prioritizes the most congested movements without central control.

We employ this auction model to implement priority-based arbitration. The auction provides a clear, structured process for deciding which movement gets the green light. Our work extends this concept by enhancing the bid calculation – instead of relying only on current sensor data, our bids are dynamically adjusted based on predicted flow data from other intersections, shared via our negotiation protocol.

3 System Model and Formulation

The traffic network is modeled as a directed graph $\mathcal{G} = (\mathcal{I}, \mathcal{E})$. For each agent $I_i \in \mathcal{I}$:

- M_i : The set of traffic movements.
- s_i^t : The local state $\{Q_{i,m}^t, W_{i,m}^t\}$ for each movement $m \in M_i$.
- ϕ_i^t : The active signal phase at time t .
- $\mathcal{N}_i$: A set of all neighbor agents for I_i .
- (k_i^{pub}, k_i^{priv}) : The key pair of agent I_i for signing messages.
- $\mathcal{T}_i$: The allow-list of agent I_i , containing $(agent_id, public_key)$ tuples of trusted neighbors and sensors.

The protocol is not particular about the source of the data. In our hardware proof-of-concept, local state variables are estimated from inductive loop detector data.

3.1 Traffic Movements

To manage the complexity of an intersection, we disaggregate traffic flow into a set of discrete traffic movements. A movement, $m \in M_i$, represents a unique, controlled path through the intersection, such as a specific left turn or a straight-through flow. It is formally defined as a tuple (origin, destination, lanes, turn), where origin and destination correspond to edges in the graph $\mathcal{G}$. This structure provides a complete abstraction for any real-world lane configuration, allowing multiple physical lanes (e.g., a dual left-turn lane group) to be modeled as a single, distinct movement. This granular representation is crucial for determining non-conflicting traffic flows and for calculating the capacity of each path.

3.2 Signal Phases and Colors

A signal phase defines a set of non-conflicting movements that are granted right-of-way simultaneously. For each agent I_i , we predefine a set of legal, non-conflicting signal phases, Φ_i . Each phase $\phi \in \Phi_i$ is a static configuration that grants a green light to a compatible set of movements while assigning a red light to all others. Formally, a phase can be represented as a bitstring $\phi \in \{0, 1\}^{|M_i|}$, where each bit corresponds to the state of a particular movement (1 for green, 0 for red). The set Φ_i contains only those combinations that are guaranteed to be safe.

The agent's decision at each step is the selection of a single phase $\phi_i^{t+1} \in \Phi_i$ to activate. The underlying signal controller is responsible for managing the state transitions between phases, including the insertion of yellow lights as appropriate.

4 Methodology

4.1 Congestion-Aware Negotiation Protocol

An agent I_i first estimates its potential for congestion, C_i^t , based on its current state and predictions received from its neighbors.

$$C_i^t = \max_{m \in M_i} \frac{\left(\sum_{j \in \mathcal{N}_i} P_{j \rightarrow i, m}^t \right) + Q_{i, m}^t}{\mathbb{K}_{i, m}} \quad (1)$$

The numerator is the total anticipated load (predicted arrivals from the previous cycle plus the current queue), and the denominator, $\mathbb{K}_{i, m}$, is the road's physical capacity constant.

If C_i^t exceeds a predefined threshold constant θ , the agent calculates and sends a reduction request, $R_{i \rightarrow j}^t$, to the responsible agent I_j . This value represents the proportional reduction in traffic flow desired by agent I_i .

$$R_{i \rightarrow j}^t = 1 - \frac{\theta}{C_i^t} \quad (2)$$

4.2 Auction-Based Arbitration

Each movement computes a base bid, $b_{i, m}^t$, representing its local priority based solely on its current state. This is essentially a weighted sum using the constant weights ω_Q and ω_W .

$$b_{i, m}^t = \omega_Q \cdot Q_{i, m}^t + \omega_W \cdot W_{i, m}^t \quad (3)$$

This base bid is then adjusted by negotiation feedback. When agent I_i receives a reduction request from a neighbor, it computes an effective reduction factor that is capped by its own operational constraints (e.g., minimum green time). This effective factor is then used to adjust the bid.

$$b_{i, m}^{t'} = b_{i, m}^t \times \left(1 - \min \left(R_{j \rightarrow i}^t, 1 - \frac{\mathbb{D}_{min}}{\mathbb{D}_{priority}} \right) \right) \quad \text{where } j = \text{Destination}(m) \quad (4)$$

Once the movement with the highest adjusted bid, m^* , is chosen to turn green, a specific signal phase must be selected. Since m^* might be green in several valid phases, a deterministic tie-breaking rule is required. The agent evaluates the utility of each candidate phase by summing the adjusted bids of all movements it would serve. The phase yielding the highest aggregate utility is selected, thereby optimizing the green light allocation not only for the winning movement but for the entire set of compatible movements.

4.3 Auction Triggering and Phase Stability

An auction is not executed at every discrete time step. To prevent rapid, unstable signal changes (flickering), the decision to hold an auction is governed by a set of time-based stability rules. Let Δt_{phase} be the time elapsed since the last phase change. We define three duration constants:

Minimum Duration. ($\mathbb{D}_{min}$) The absolute minimum time a phase must remain active for safety and driver expectation.

Priority Duration. ($\mathbb{D}_{priority}$) A period during which the currently active phase has priority.

Release Duration. ($\mathbb{D}_{release}$) A maximum duration after which the current phase loses its priority and is subject to being replaced.

The logic for triggering an auction is as follows:

1. If $\Delta t_{phase} < \mathbb{D}_{min}$, the current phase ϕ_i^t is maintained. No auction is held.
2. If $\mathbb{D}_{min} \leq \Delta t_{phase} < \mathbb{D}_{priority}$, an auction is triggered only if the total bid for the current phase, $B(\phi_i^t) = \sum_{m \text{ green in } \phi_i^t} b'_{i,m}$, becomes negative. Otherwise, ϕ_i^t is maintained.
3. If $\mathbb{D}_{priority} \leq \Delta t_{phase} < \mathbb{D}_{release}$, an auction is triggered if any other phase $\phi' \in \Phi_i$ has a positive total bid, $B(\phi') > 0$, creating competition.
4. If $\Delta t_{phase} \geq \mathbb{D}_{release}$, a **handicapped auction** is held. In this auction, the current phase ϕ_i^t is disqualified (its bid is treated as zero or negative infinity) to ensure that a phase change is considered, preventing indefinite green time for one direction.

4.4 Future Traffic Flow Prediction

The selected phase, ϕ_i^{t+1} , is now determined. The agent then predicts its future impact, $P_{i \rightarrow j,m}^{t+1}$, for each neighbor I_j . This prediction is conditioned on both the current state s_i^t and the selected phase ϕ_i^{t+1} . The resulting predictions are broadcast to neighbors for their use in the next decision cycle.

4.5 Secure Communication via PKI

All agent-to-agent communication is secured using public-key cryptography. Every message is signed by the sender I_i using its private key, k_i^{priv} . A recipient I_j verifies the message against its local allow-list, $\mathcal{T}_j$. A message is accepted only if the sender is on the list and the signature is valid. This prevents tampering and spoofing attacks.

4.6 Resilience to Network Disruption

Let $\mathcal{A}_i^t$ be the set of neighbors from whom agent I_i has received a valid, authenticated message within a timeout window $\Delta \mathbb{T}_H$.

Congestion Calculation. Congestion is calculated using an effective predicted inflow, $\hat{P}$, which falls back to a historical average, $\bar{P}$, for inactive neighbors.

$$\hat{C}_i^t = \max_{m \in M_i} \frac{\left(\sum_{j \in \mathcal{N}_i} \hat{P}_{j \rightarrow i,m}^t \right) + Q_{i,m}^t}{\mathbb{K}_{i,m}} \quad \text{where} \quad \hat{P}_{j \rightarrow i,m}^t = \begin{cases} P_{j \rightarrow i,m}^t & \text{if } I_j \in \mathcal{A}_i^t \\ \bar{P}_{j \rightarrow i,m} & \text{if } I_j \notin \mathcal{A}_i^t \end{cases} \quad (5)$$

Bid Adjustment. Bid adjustment only considers feedback from active neighbors, applied selectively.

$$b_{i,m}''' = b_{i,m}^t \times \left(1 - \min \left(R_{j \rightarrow i}^t, 1 - \frac{\mathbb{D}_{min}}{\mathbb{D}_{priority}} \right) \right) \quad \text{where } j = \text{Destination}(m) \text{ and } j \in \mathcal{A}_i^t \quad (6)$$

If $\mathcal{A}_i^t$ is empty, these equations naturally degrade to a local-only adaptive controller.

5 Algorithm

The following algorithm is the product of the discussions in the previous sections.

Algorithm 1 Control Loop for Intersection Agent I_i

Require: State s_i^t , Current phase ϕ_i^t , Time since last phase change Δt_{phase} .

Ensure: Next phase ϕ_i^{t+1} .

- 1: $b' \leftarrow \text{CALCULATEADJUSTEDBIDS}(s_i^t)$
 - 2: $\phi_i^{t+1} \leftarrow \text{DETERMINENEXTPHASE}(\phi_i^t, \Delta t_{\text{phase}}, b', \Phi_i)$
 - 3: $\text{PREDICTANDBROADCAST}(s_i^t, \phi_i^{t+1})$
 - 4:
 - 5: **return** ϕ_i^{t+1}
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Algorithm 2 Supporting Functions

- 1: **function** $\text{CALCULATEADJUSTEDBIDS}(s_i^t)$
 - 2: Process incoming messages to update active neighbor set $\mathcal{A}_i^t$.
 - 3: **for all** movement $m \in M_i$ **do**
 - 4: Calculate adjusted bid $b'_{i,m}$ from local state and requests from active neighbors.
 - 5: **end for**
 - 6: **return** adjusted bids b'
 - 7: **end function**

 - 8: **function** $\text{DETERMINENEXTPHASE}(\phi_i^t, \Delta t_{\text{phase}}, b', \Phi_i)$
 - 9: **if** $\Delta t_{\text{phase}} < \mathbb{D}_{\min}$ **then**
 - 10: **return** ϕ_i^t
 - 11: **else if** $\Delta t_{\text{phase}} < \mathbb{D}_{\text{priority}}$ **then**
 - 12: **return** ϕ_i^t **unless** $\sum_{m \in \phi_i^t} b'_{i,m} < 0$ **then** $\text{RUNAUCTION}(\Phi_i, b')$
 - 13: **else if** $\Delta t_{\text{phase}} < \mathbb{D}_{\text{release}}$ **then**
 - 14: **return** ϕ_i^t **unless** $\exists \phi' \in \Phi_i \setminus \{\phi_i^t\} : \sum_{m \in \phi'} b'_{i,m} > 0$ **then** $\text{RUNAUCTION}(\Phi_i, b')$
 - 15: **else**
 - 16: **return** $\text{RUNAUCTION}(\Phi_i \setminus \{\phi_i^t\}, b')$
 - 17: **end if**
 - 18: **end function**

 - 19: **function** $\text{RUNAUCTION}(\Phi_{\text{candidates}}, b')$
 - 20: $m^* \leftarrow \arg \max_m \{b'_{i,m} \mid m \text{ can be served by some } \phi \in \Phi_{\text{candidates}}\}$.
 - 21: $\Phi_{\text{cand}} \leftarrow \{\phi \in \Phi_{\text{candidates}} \mid m^* \text{ is green in } \phi\}$.
 - 22: **return** $\arg \max_{\phi \in \Phi_{\text{cand}}} \sum_{m \text{ is green in } \phi} b'_{i,m}$.
 - 23: **end function**

 - 24: **function** $\text{PREDICTANDBROADCAST}(s_i^t, \phi_i^{t+1})$
 - 25: Predict the planned outflow P^{t+1} using the current state and phase (s_i^t, ϕ_i^{t+1}) .
 - 26: Sign and broadcast message with P^{t+1} to neighbors.
 - 27: **end function**
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6 Simulation and Hardware Implementation

6.1 Simulation with SUMO

The proposed system will be validated using the Simulation of Urban MObility (SUMO) suite. We will define a multi-intersection road network and realistic traffic demand using SUMO’s standard XML formats. The agent logic will be implemented in Python and will interface with the running simulation. At each simulation step, the script will retrieve the simulated sensor data, execute the agent algorithms, and send signal control commands back to SUMO.

A simple setup for the simulation could involve 5 intersections arranged in a plus shape, with the peripheral intersections acting as T-junctions.

6.2 Hardware Proof-of-Concept

A physical prototype will demonstrate the feasibility of the edge agent concept.

- Inductive loop sensors to detect cars, and if possible, a radar-based sensor to detect their speed – this could be another parameter used to compute the bid.
- An ESP32 microcontroller will interface with the sensors, processing the raw data and publishing the results over the network (e.g., via MQTT) or send it directly to the core agent via bluetooth.
- Another ESP32 will serve as the intersection agent, subscribing to the sensor data and executing the main control logic - the prediction, negotiation and the auction.
- Communication between agents at intersections will occur over Wi-Fi or LAN, similar to how signal mounted CCTV cameras communicate with a central control room. The agent’s decisions will be visualized by controlling an LED array that mimics a traffic light.

7 Evaluation Metrics and Experiments

7.1 Metrics

The system’s performance will be evaluated using four key metrics to provide a comprehensive view of its impact on both individual driver experience and overall network efficiency.

Average Travel Time. Measures the full origin-to-destination journey time, serving as the primary indicator of overall system efficiency and user experience. It will be used to quantify the performance gains over the fixed-time baseline and to isolate the specific value contributed by the prediction and negotiation components in the experiments.

Average Delay. Isolates the time vehicles spend stopped or slowed at intersections, directly measuring the performance of the auction and negotiation mechanisms. This metric helps verify that local congestion is being resolved effectively, rather than simply being shifted to a neighboring intersection.

Average Queue Length. Quantifies the number of vehicles waiting at an intersection, providing a direct measure of localized congestion and the risk of gridlock-inducing spillback. This is helpful for evaluating the system’s stability in the *Heavy Congestion* scenario.

Throughput. Measures the total number of vehicles that complete their trips per hour, evaluating the network’s overall capacity. This ensures that improvements in individual travel times do not degrade the system’s ability to handle high traffic volume, which is needed to assess the resilience of the system in the *Network Disruption* experiment.

7.2 Experiments

Name	Description
Fixed-Time Control	A standard, non-adaptive controller. Provides a baseline.
Without Prediction	System run with $P^{t+1} = 0$. Isolates the value add of prediction.
Without Negotiation	System run with $R^t = 0$. Isolates the value add of negotiation.
Network Disruption	Simulate communication link failures to evaluate the graceful degradation mechanisms and recovery speed.
Adversarial Attack	Simulate a rogue node injecting malicious messages to validate that the agent rejects all unauthorized messages.
Heavy Congestion	Simulate a scenario where almost all traffic is directed towards one destination.
Emergency Vehicles	Simulate a scenario where an emergency vehicle requires right-of-way.

References

- [1] Jiang, Q., et al. (2022). Multi-Agent Reinforcement Learning for Traffic Signal Control through Universal Communication Method. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)*.
- [2] Baluja, S., Covell, M., & Sukthankar, R. (2017). Traffic Lights with Auction-Based Controllers: Algorithms and Real-World Data. *arXiv preprint arXiv:1702.01205*.